

Learning Under Non-Stationarity: Statistical Inference & Bandit Convex Optimization

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Algorithms and Computationally Intensive Inference Seminar
Department of Statistics, University of Warwick

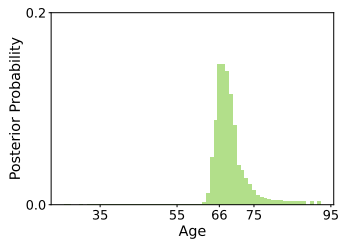
13 March 2026

High-dimensional
changepoint
localization

High-dimensional changepoint localization

Offline inference:

- 👁️ Observe patients' medical features & heart failure outcomes
- 🕒 Relationship between features & outcome **change** markedly at some age
- ❓ At what age does change happen?



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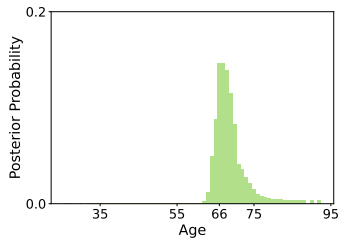
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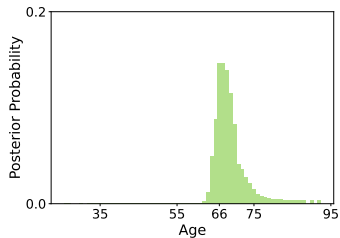
Non-stationary
bandit convex
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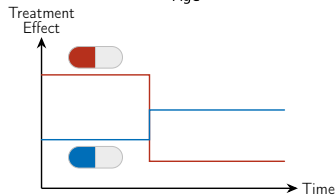
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Adaptive decision-making:

- 👁️ Observe outcomes *only* for the chosen treatment
- 🕒 Treatment effects **change** over time
- ? How to balance exploration-exploitation?



**Statistical
aim**



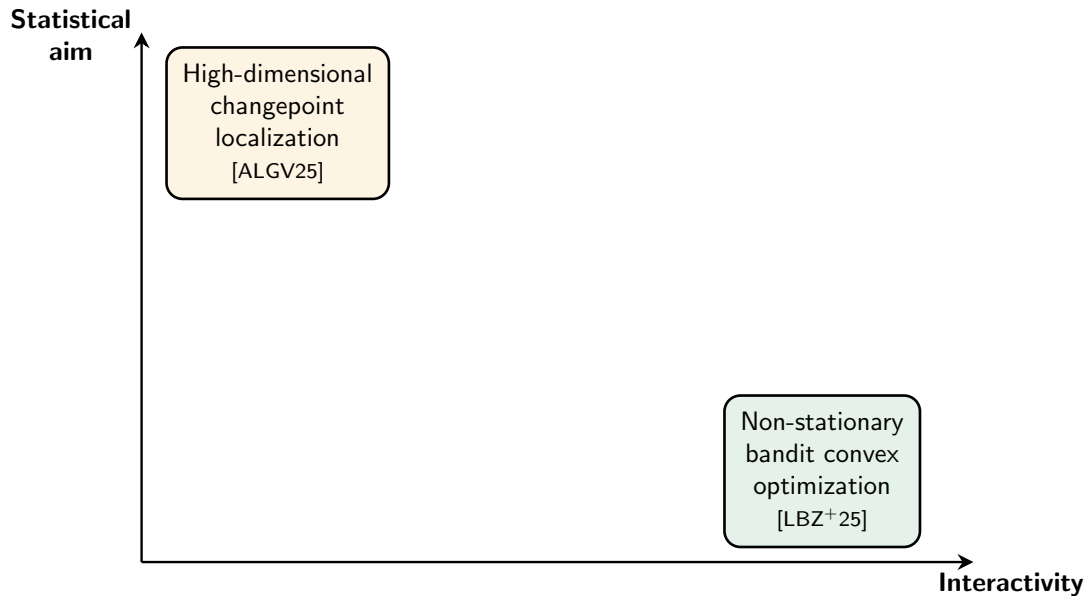
A blank 2D coordinate system is shown. The vertical axis (y-axis) is labeled 'Statistical aim' at its top. The horizontal axis (x-axis) is labeled 'Interactivity' at its right end. Both axes are represented by black lines with arrowheads at their respective ends. The axes intersect at an origin point in the bottom-left corner of the plot area.

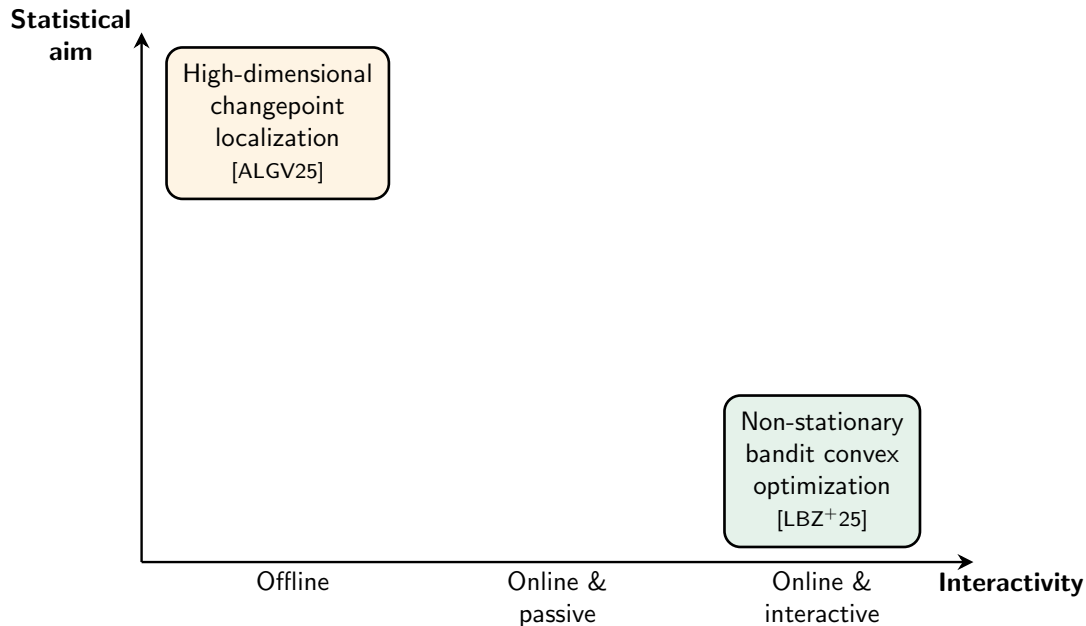
Interactivity

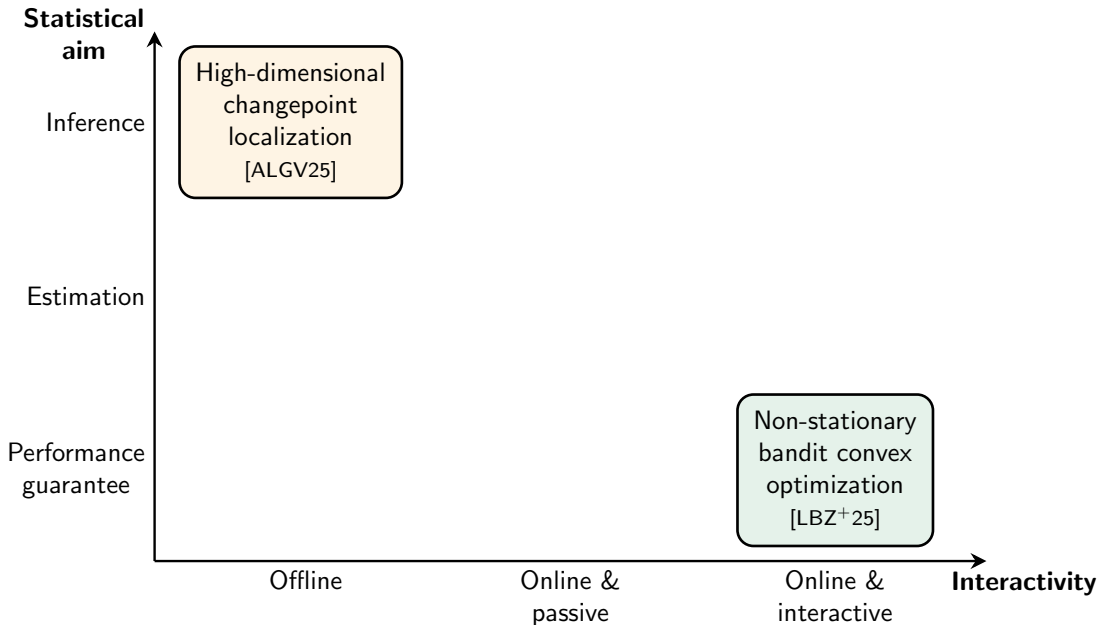
**Statistical
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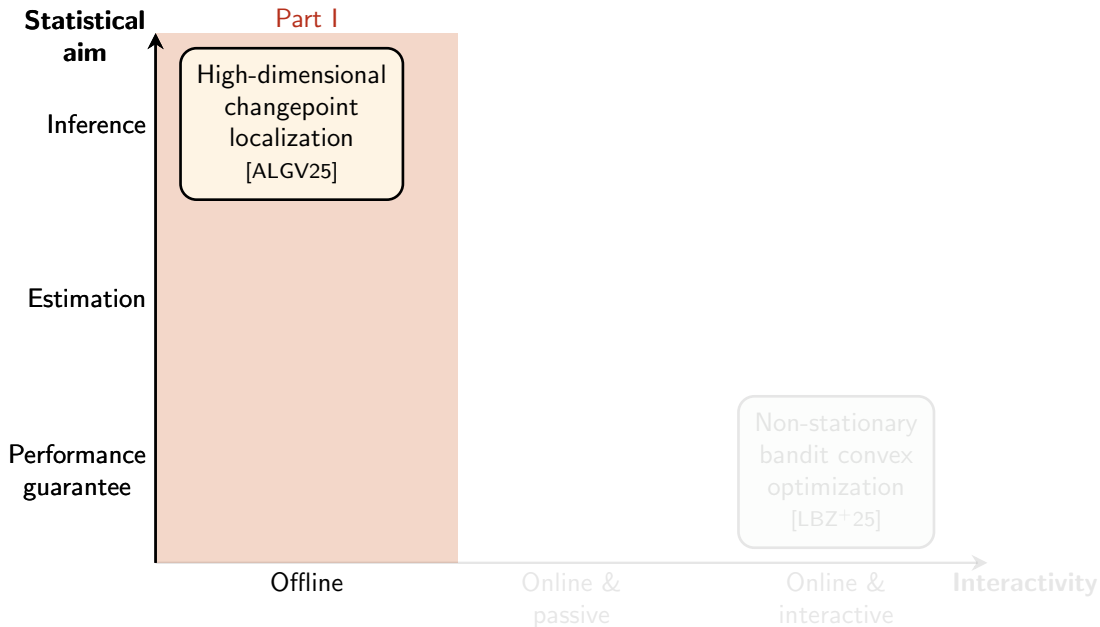
High-dimensional
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[ALGV25]

Interactivity









High-dimensional changepoint inference

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For $i = 1, 2, \dots, n$,
 $\mathbf{x}_i, \boldsymbol{\beta}^{(\ell)} \in \mathbb{R}^d$

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta}^{(1)} + \varepsilon_i$$

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$i = \eta_1$

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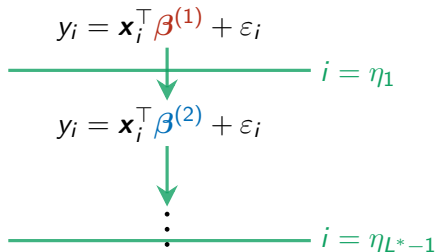
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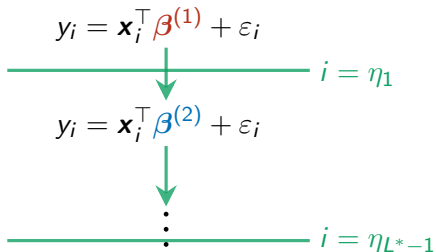
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L^* is unknown constant,
with $L^* < L$

$\boldsymbol{\eta} := [\eta_1, \dots, \eta_{L^*-1}]$,
with $\eta_{\ell+1} - \eta_\ell \asymp n$

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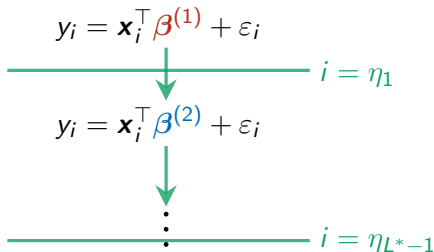
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Model: entries of $\mathbf{x}_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \frac{1}{n})$, entries of $\boldsymbol{\beta}^{(\ell)} \stackrel{\text{i.i.d.}}{\sim} p_{\bar{\boldsymbol{\beta}}}$

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Goal: infer changepoint locations $(\eta_\ell)_{\ell=1}^{L^*-1}$ from $(\mathbf{x}_i, y_i)_{i=1}^n$, and estimate signals $(\boldsymbol{\beta}^{(\ell)})_{\ell=1}^{L^*}$

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Scaling regime: $n, d \rightarrow \infty$ with $n/d \rightarrow$ fixed constant δ

For $d \ll n$,

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Existing estimation methods:

- For sparse $p_{\tilde{\beta}}$: **partition** + **penalized maximum likelihood** [LB16, RWW⁺21, BS23]
- For sparse-difference $p_{\tilde{\beta}}$: complementary sketching [GW22]
- For non-sparse $p_{\tilde{\beta}}$: [CKL25, LQZL26]

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Based on **Approximate Message Passing**
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- 👍 Allow general $p_{\tilde{\beta}}$; credible sets with precise asymptotic characterization
- 👎 No minimax optimal localization error

Aside: AMP for sparse linear regression

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$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

entries of $\mathbf{X} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \frac{1}{n})$

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modified residual: $\mathbf{r}_t = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}_t + b_t \mathbf{r}_{t-1}$

signal estimate: $\boldsymbol{\beta}_{t+1} = f_{\text{soft thres}}(\boldsymbol{\beta}_t + \mathbf{X}^\top \mathbf{r}_t; \theta_t)$

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$$y = X\beta + \varepsilon$$

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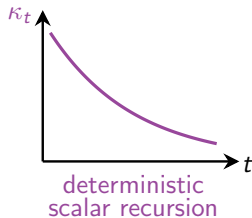
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$$\Theta_t = \mathbf{X}\hat{\mathbf{B}}_t - \mathbf{R}_{t-1}\mathbf{F}_t^\top$$

$$\mathbf{R}_t = g_t(\Theta_t, \mathbf{y})$$

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Theorem (informal) [ALGV25].

$$\text{rows of } \mathbf{B}_t - \mathbf{B}\nu_{\mathbf{B},t} \approx \mathbf{G}_{\mathbf{B},t}$$

$$\mathbf{G}_{\mathbf{B},t} \sim \mathcal{N}(\mathbf{0}, \kappa_{\mathbf{B},t})$$

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g_t uses prior info
about change-
points $\eta \sim p_{\bar{\eta}}$

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$$\text{rows of } \mathbf{B}_t - \mathbf{B} \nu_{\mathbf{B},t} \approx \mathbf{G}_{\mathbf{B},t}$$

$$\mathbf{G}_{\mathbf{B},t} \sim \mathcal{N}(\mathbf{0}, \kappa_{\mathbf{B},t})$$

$$\text{rows of } \Theta_t - \Theta \nu_{\Theta,t} \approx \mathbf{G}_{\Theta,t}$$

$$\mathbf{G}_{\Theta,t} \sim \mathcal{N}(\mathbf{0}, \kappa_{\Theta,t})$$

AMP for Bayesian changepoint inference

Signal is $\mathbf{B} = \begin{bmatrix} \uparrow & & \uparrow \\ \beta^{(1)} & \dots & \beta^{(L)} \\ \downarrow & & \downarrow \end{bmatrix} \overset{\text{rows}}{\underset{\text{i.i.d.}}{\sim}} p_{\bar{\mathbf{B}}}$ and define $\Theta := \mathbf{X}\mathbf{B} \overset{\text{rows}}{\underset{\text{i.i.d.}}{\sim}} p_{\bar{\Theta}}$.

g_t uses prior info
about change-
points $\eta \sim p_{\bar{\eta}}$

$$\Theta_t = \mathbf{X}\hat{\mathbf{B}}_t - \mathbf{R}_{t-1}\mathbf{F}_t^\top$$

$$\mathbf{R}_t = g_t(\Theta_t, \mathbf{y})$$

produce residual $\mathbf{R}_t$,
infer changepoints

$$\mathbf{B}_{t+1} = \mathbf{X}^\top \mathbf{R}_t - \hat{\mathbf{B}}_t \mathbf{C}_t^\top$$

$$\hat{\mathbf{B}}_t = f_t(\mathbf{B}_t) = \mathbb{E}[\bar{\mathbf{B}} \mid \bar{\mathbf{B}}\nu_{\mathbf{B},t} + \mathbf{G}_{\mathbf{B},t} = \mathbf{B}_t]$$

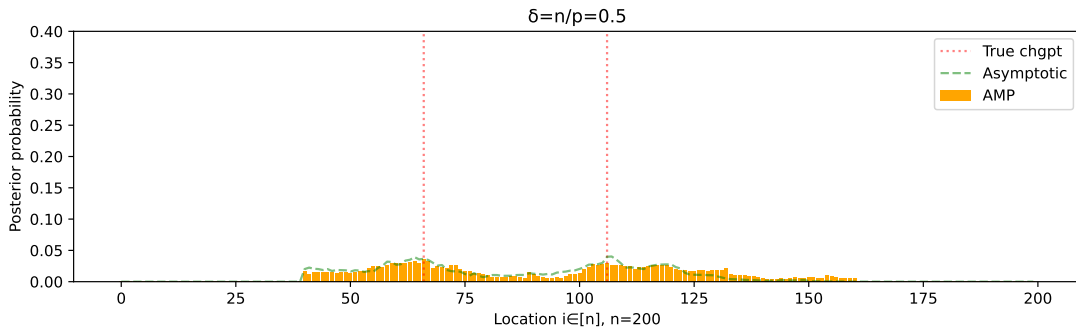
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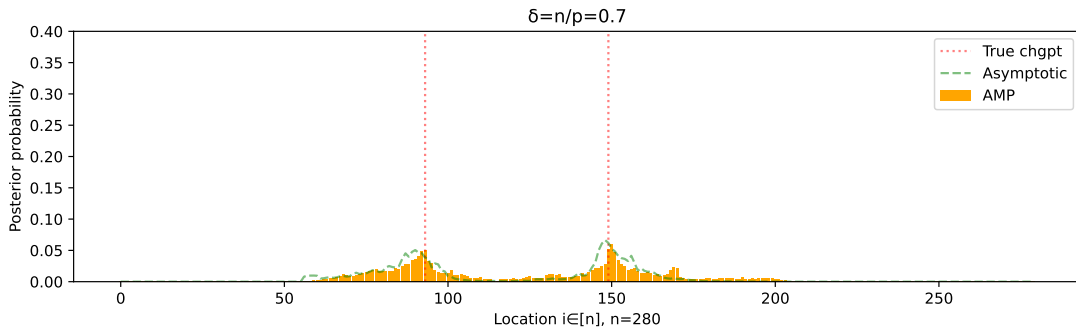
$$p(\eta \mid \Theta_t, \mathbf{y}) \stackrel{P}{\simeq} p(\eta \mid \bar{\Theta}\nu_{\Theta,t} + \mathbf{G}_{\Theta,t}, \bar{\mathbf{y}}(\bar{\Theta}, \bar{\eta}, \bar{\varepsilon}))$$

Simulations

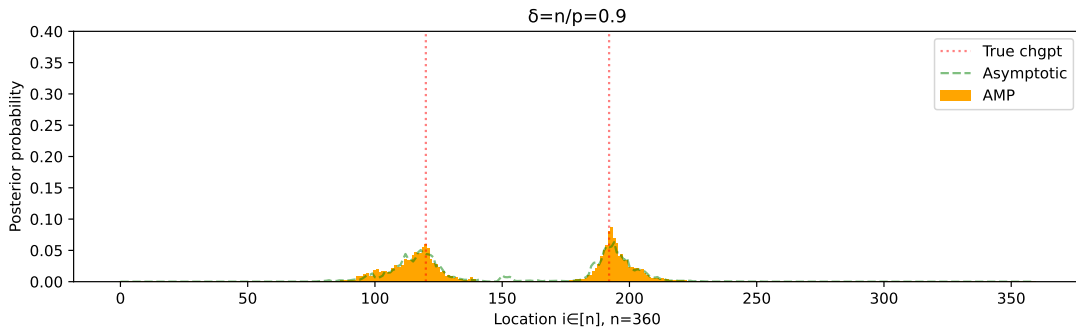
Simulations



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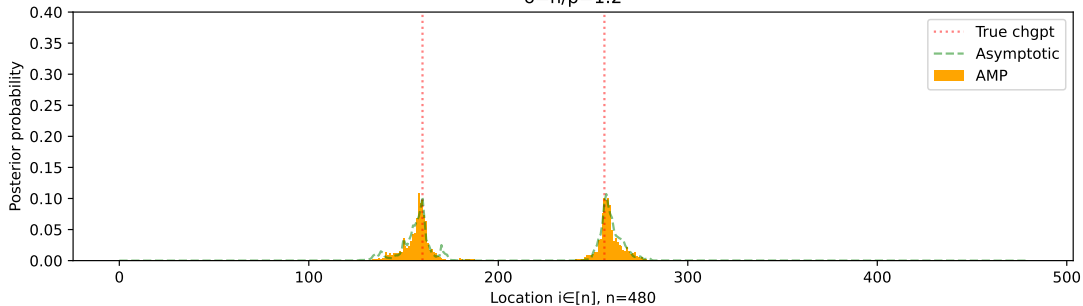


Simulations

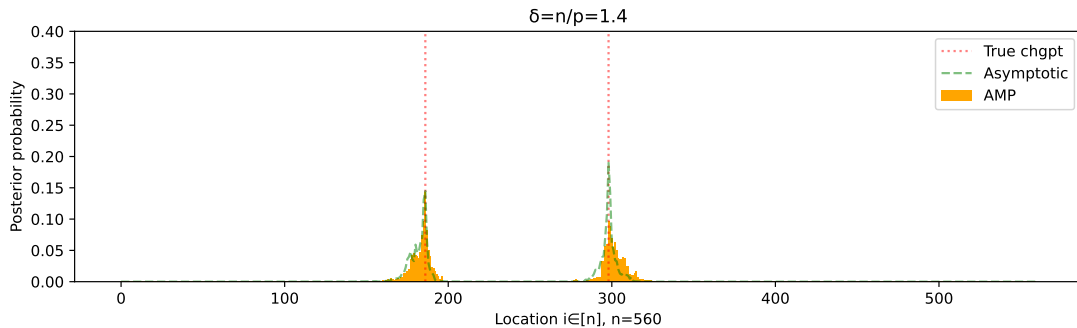


Simulations

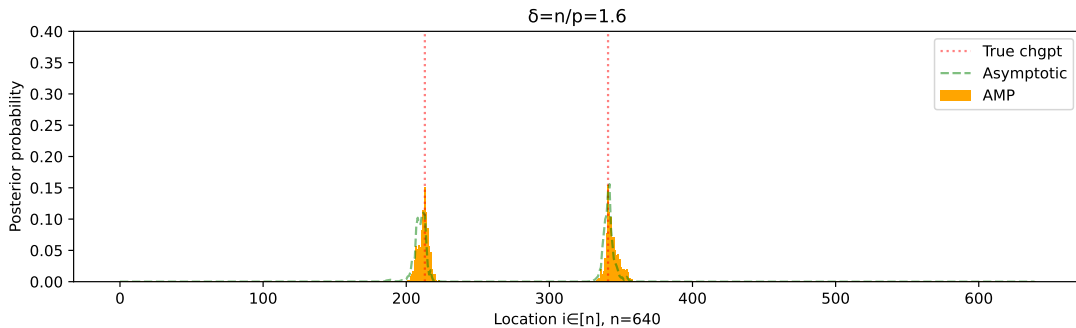
$$\delta = n/p = 1.2$$



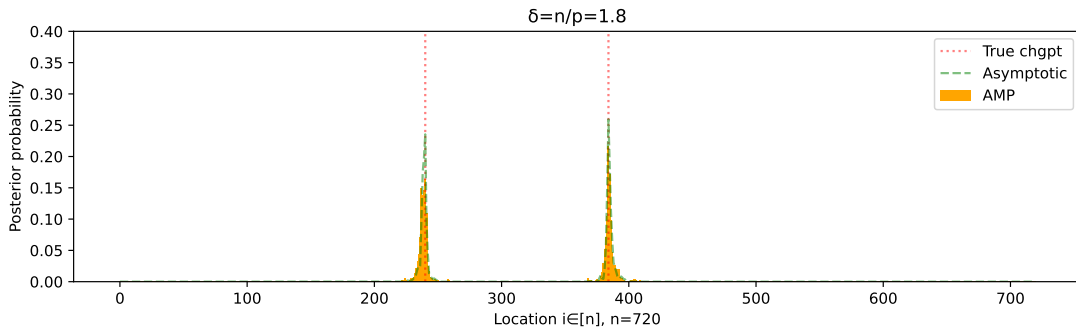
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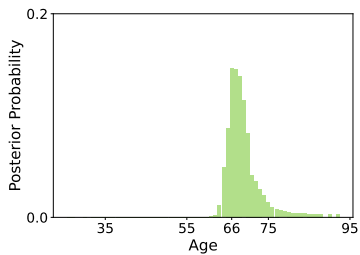


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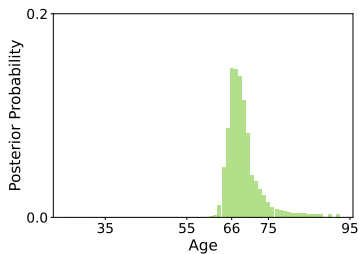
Simulations

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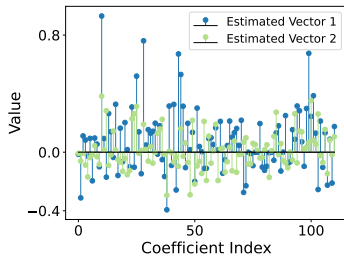


(a) Changepoint posterior vs. age.

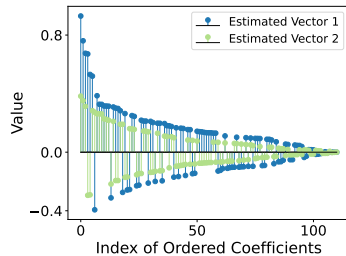
Simulations



(a) Changepoint posterior vs. age.



(b) Regression vectors.

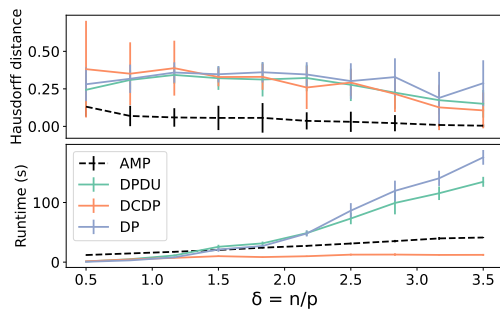


(c) Ordered regression vectors.

Figure: AMP applied to a Myocardial Infarction dataset from [GSR⁺20].

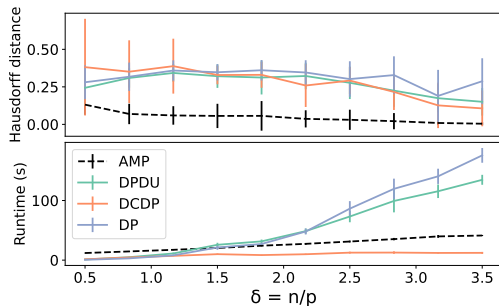
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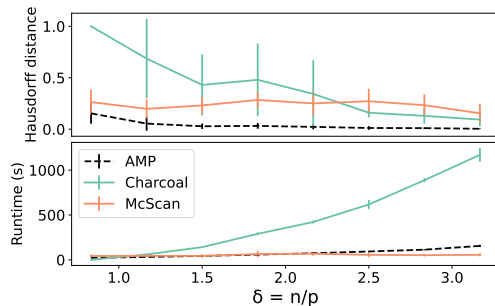


(a) Comparison with DPDU, DCDP and DP for **sparse** prior $p_{\bar{B}} = 0.5\mathcal{N}(\mathbf{0}, \delta \mathbf{I}) + 0.5\delta_0$.

Simulations

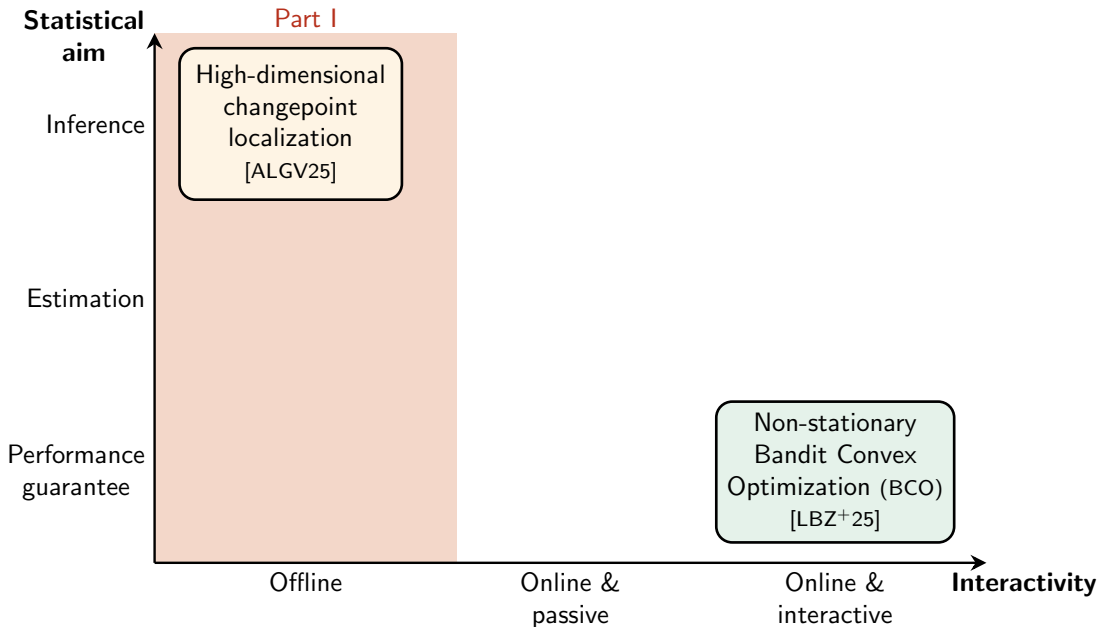


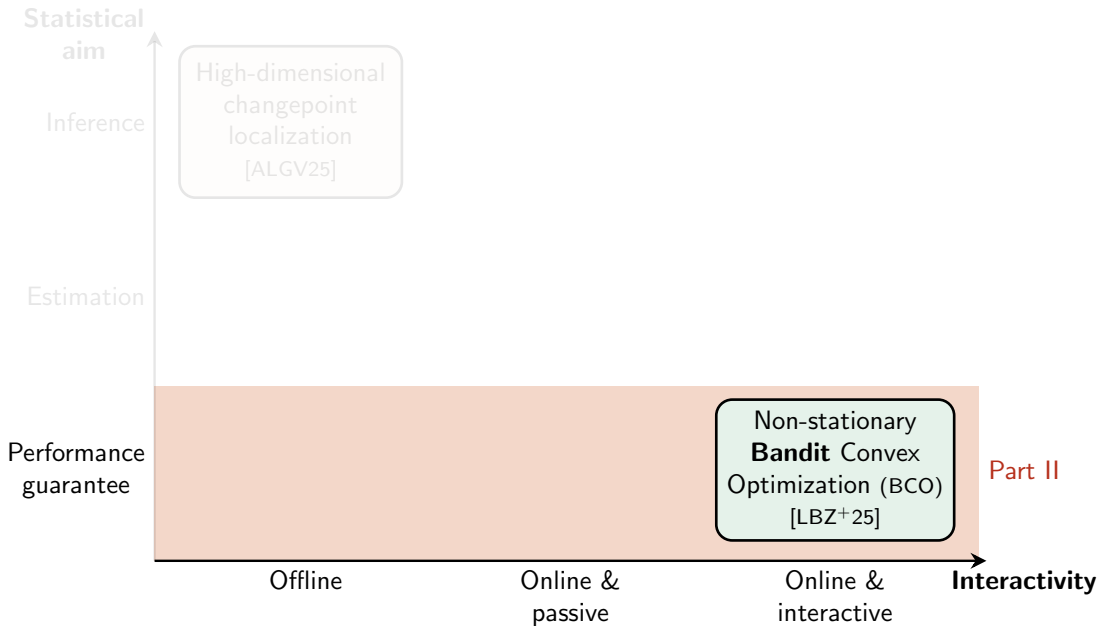
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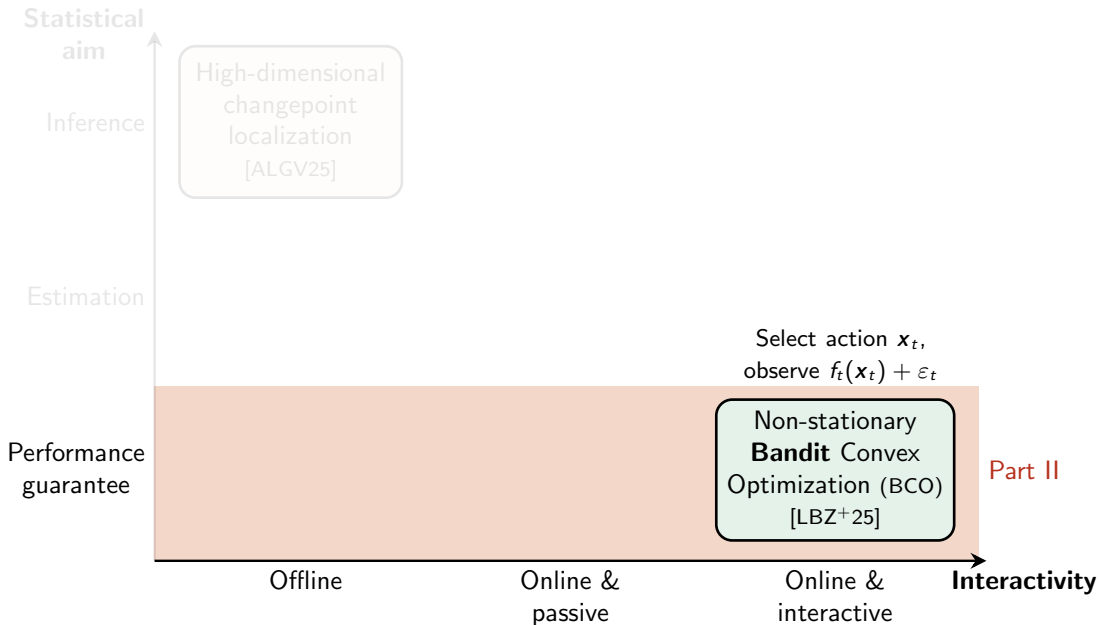


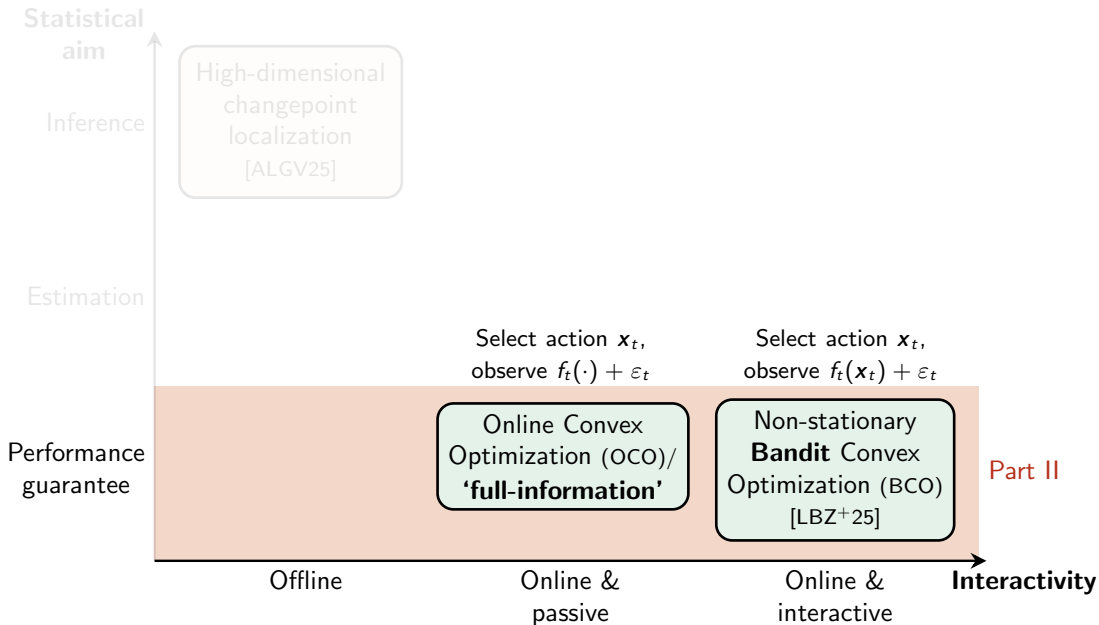
(b) Comparison with charcoal and McScan for a **sparse-difference** prior with sparsity level 0.5.

Figure: Linear model, $L^* = L = 3$.









Bandit Convex Optimization (BCO)

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- Adversary fixes **convex** loss functions $f_1, f_2, \dots, f_T : \mathbb{R}^d \rightarrow \mathbb{R}$
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$$f_t(\mathbf{x}_t) + \varepsilon_t$$

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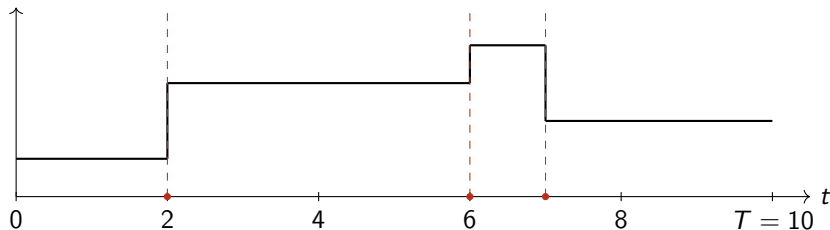
Example:

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Example: $\mathbf{u}_t$

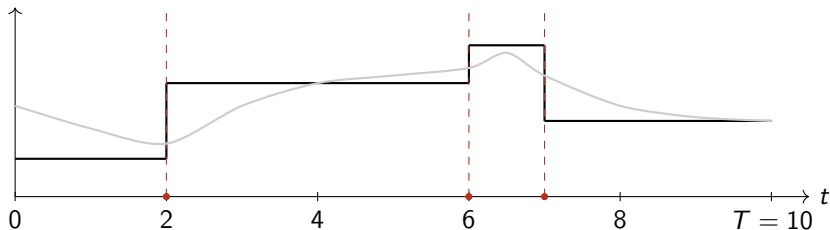


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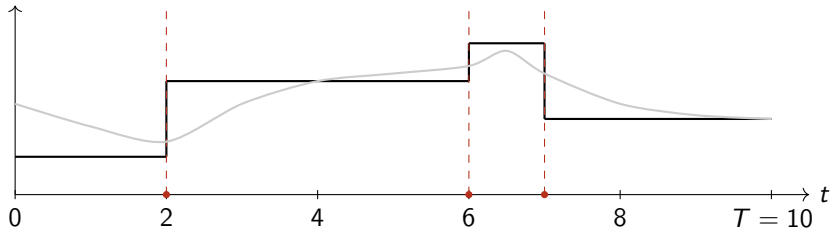


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Reduction to **adaptive** regret:

$$R^{\text{ada}}(B, T) = \max_{\substack{p, q \in [T] \\ 0 < q - p \leq B}} \max_{\mathbf{u} \in \Theta} \sum_{t=p}^q \mathbb{E}[f_t(\mathbf{x}_t) - f_t(\mathbf{u})].$$

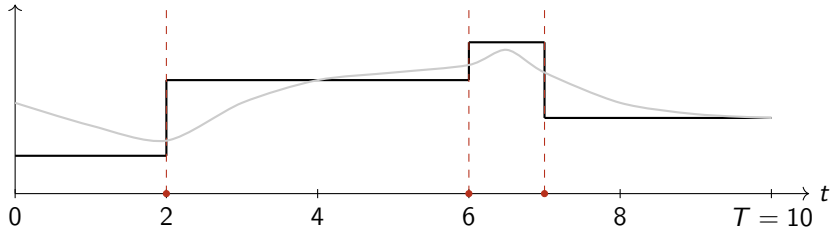
[HS09, DGSS15, AKCV16, JOWW17, Cut20, WZZ18, ZXZZ22, ZWT⁺21, ZLZ18, YWZZ24]

Non-stationary regrets

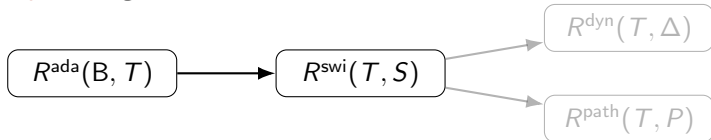
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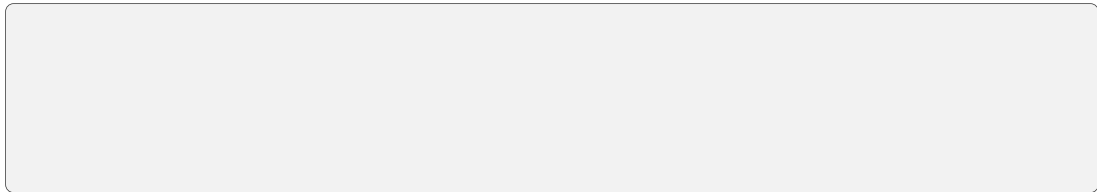


Reduction to **adaptive** regret:



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General convex (GC)	$d^{\frac{5}{2}} \sqrt{ST}$	$\sqrt{d} S^{\frac{1}{4}} T^{\frac{3}{4}}$
Strongly convex (SC)		$\underline{d\sqrt{ST}}$

Underline: **minimax-optimal** rates [LBZ⁺25].

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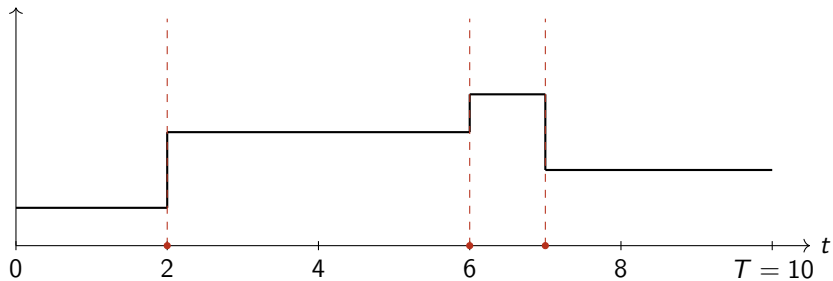
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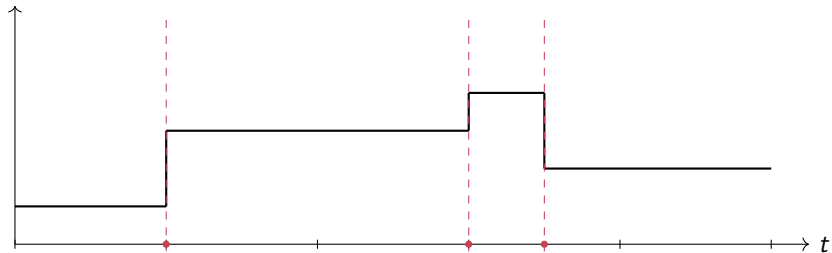
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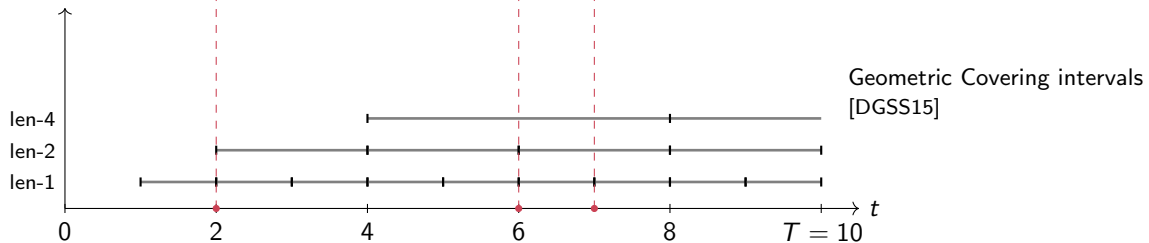
Comparator u_t



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Experts



Sleeping experts (full-information)

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for $t = 1, 2, \dots, T$ **do**

Let $\{E_1, \dots, E_{n_t}\}$ be the active experts at t with $E_i \equiv E_i(l_i)$.

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end for

Play meta-action using exponential-weights (EW):

$$\mathbf{x}_t = \sum_{i=1}^{n_t} \frac{e^{-L_{t-1,l_i}}}{\sum_{j=1}^{n_t} e^{-L_{t-1,l_j}}} \mathbf{x}_{t,l_i}$$

Observe **entire** loss function $f_t(\cdot)$

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$$\Rightarrow R^{\text{ada}}(B, T) \lesssim \begin{cases} \sqrt{B} & \text{(GC)} \\ \frac{1}{\alpha} \log B & \text{(SC)} \end{cases}.$$

[DGSS15, JOWW17, ZYZ⁺18]

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Observe **bandit feedback** $f_t(\mathbf{x}_t) + \varepsilon_t$

for $i = 1, 2, \dots, n_t$ **do**

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Play meta-action using EW: $\zeta_t \sim$ uniform on unit sphere

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Observe **bandit feedback** $f_t(\mathbf{x}_t) + \varepsilon_t$

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How to design $\ell_t(\cdot)$ using $\mathbf{g}_t$?

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- (i) Absorbs the variance of $\mathbf{g}_t$
- (ii) Together with η , adapts to unknown curvature α

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⇓ by convexity of $\hat{f}_t$

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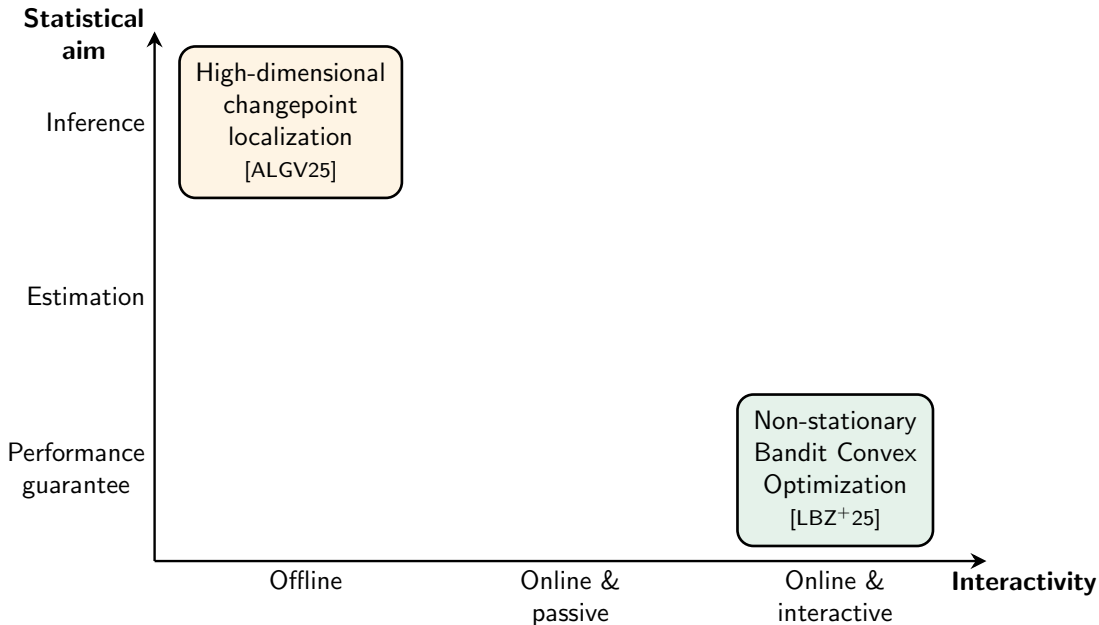
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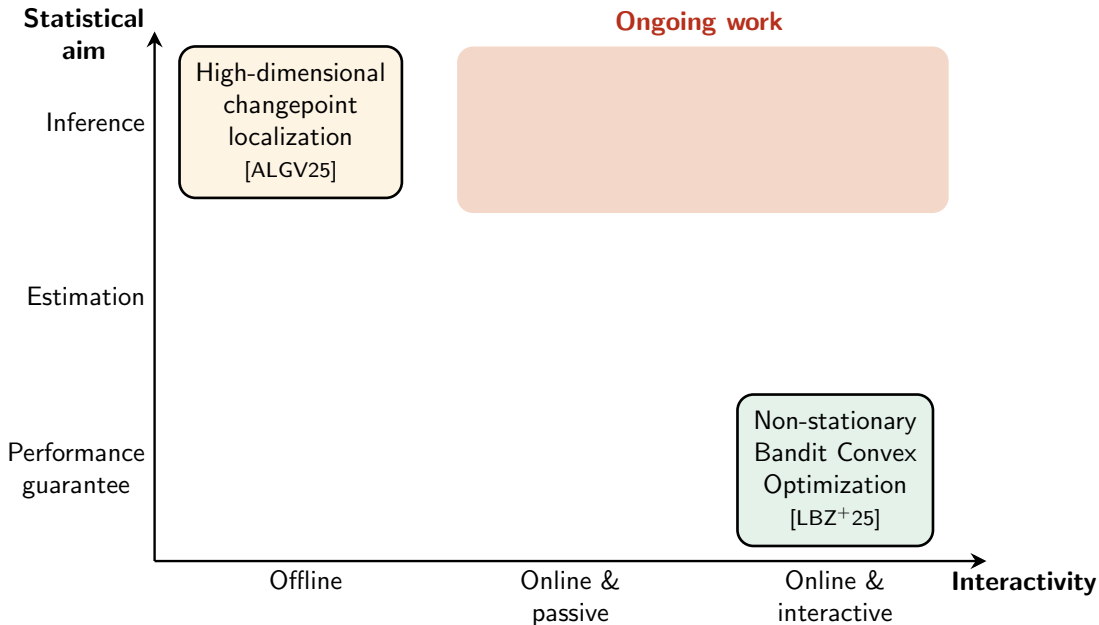
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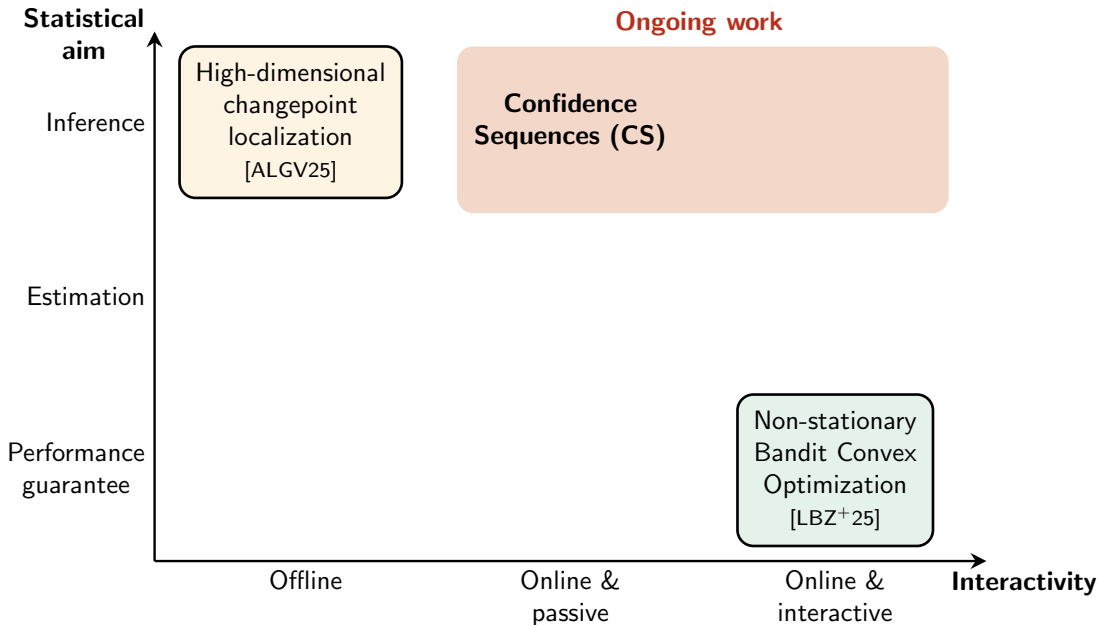
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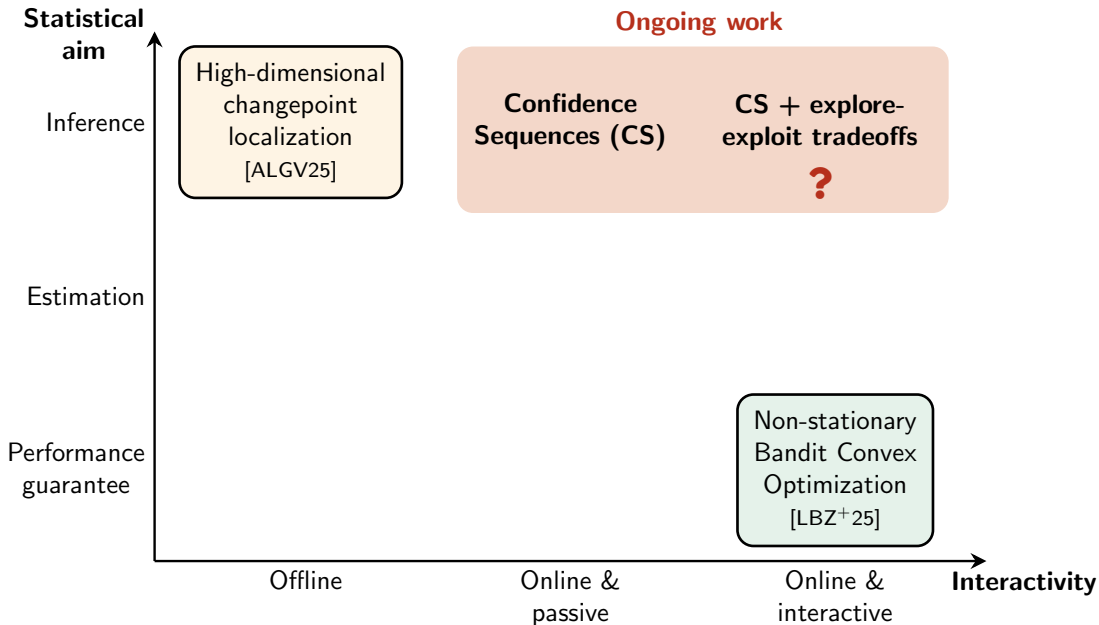
- Improve TEWA-SE through coin betting [JOWW17]?
- Leverage **second-order** information like online Newton methods from [FvdHLM24, SSNH24] that achieve state-of-the-art $\sqrt{T}$ static regrets?

Summary & ongoing work

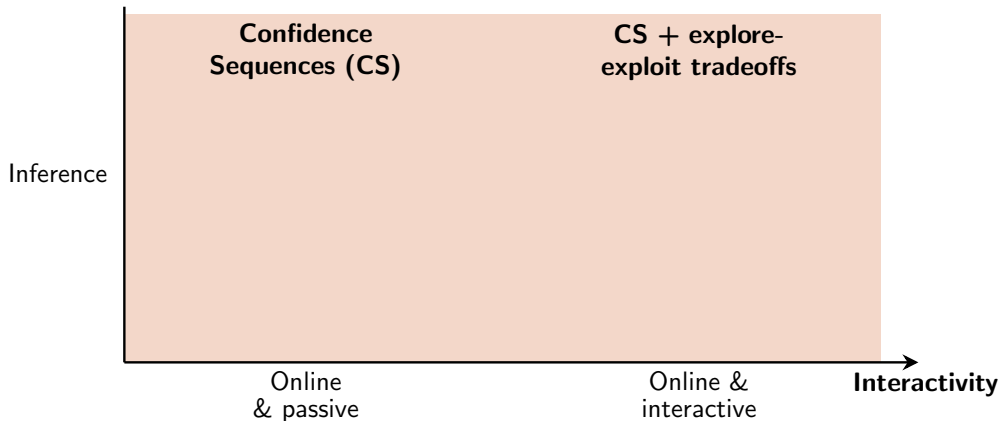






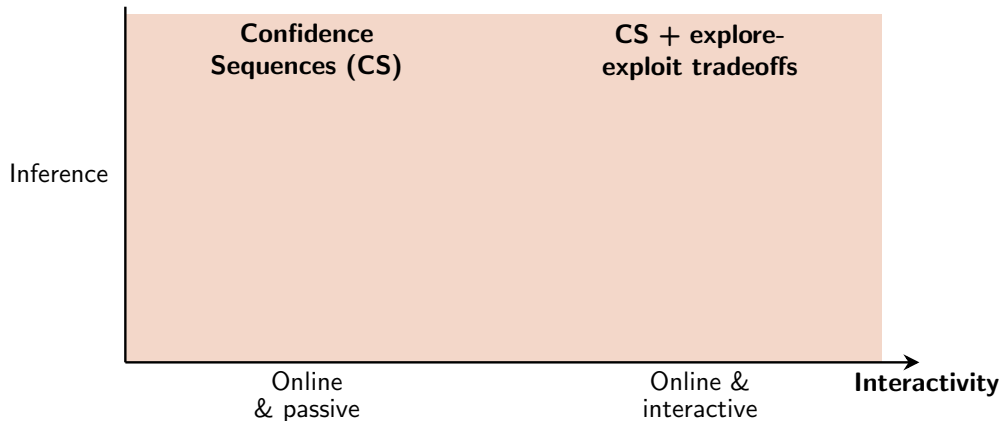


Anytime-valid **inference** for **interactive** decision-making



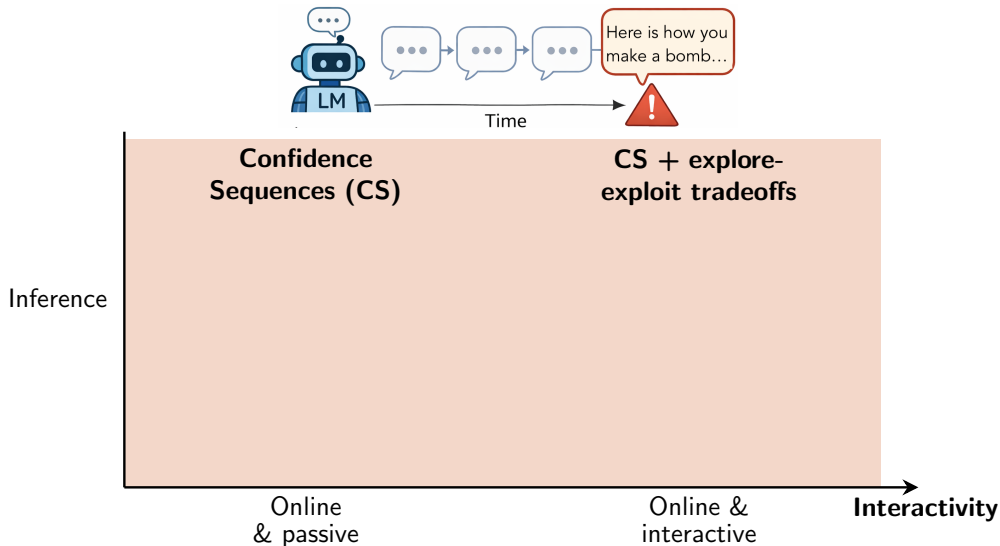
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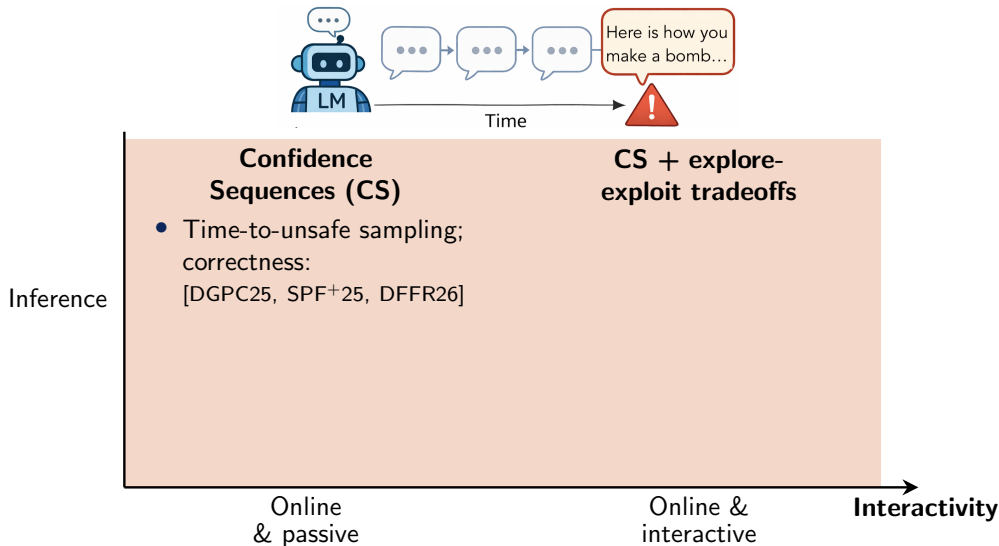
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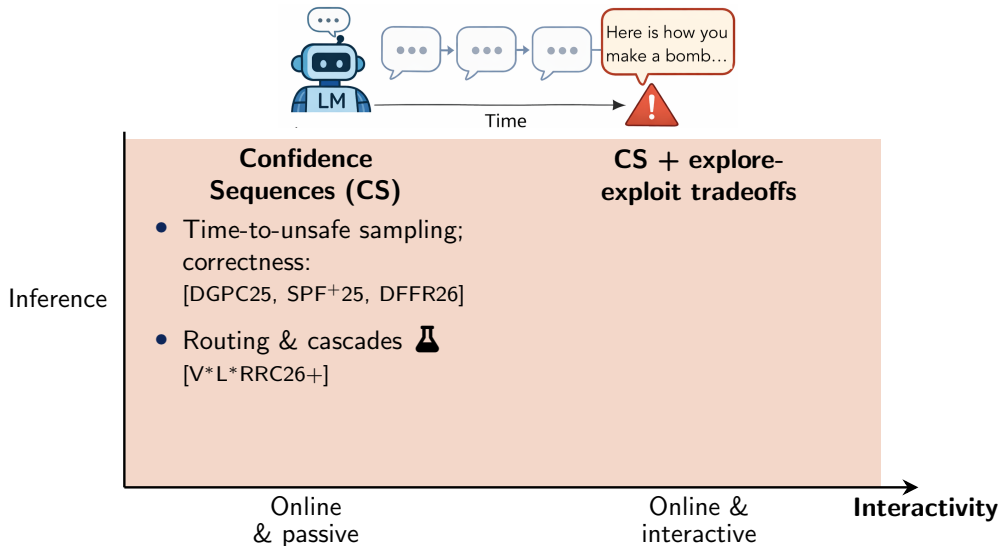
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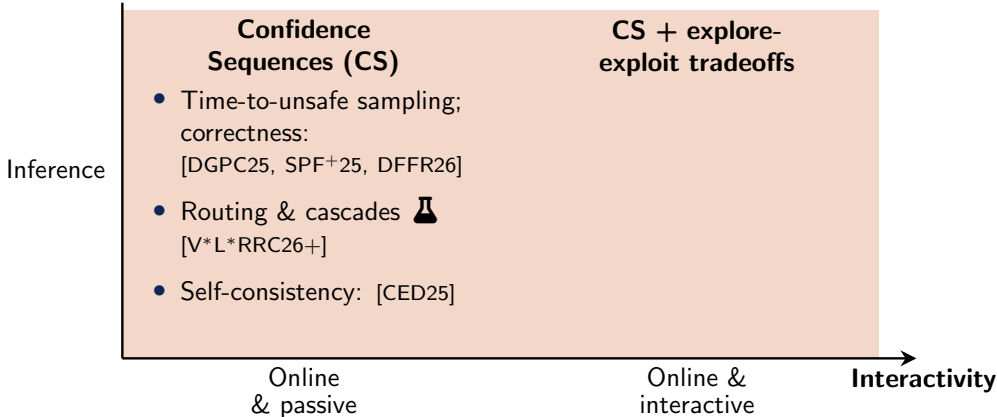
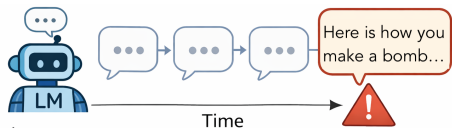
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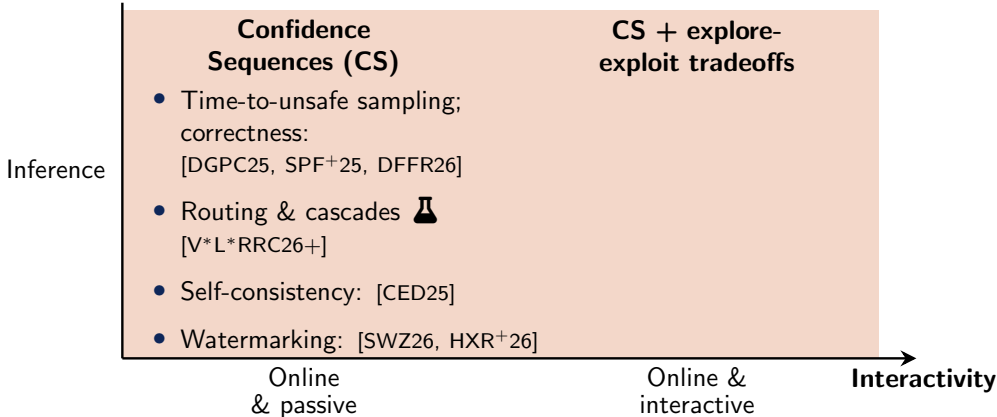
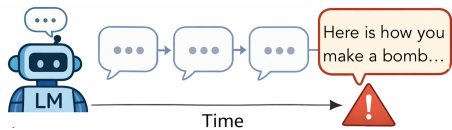
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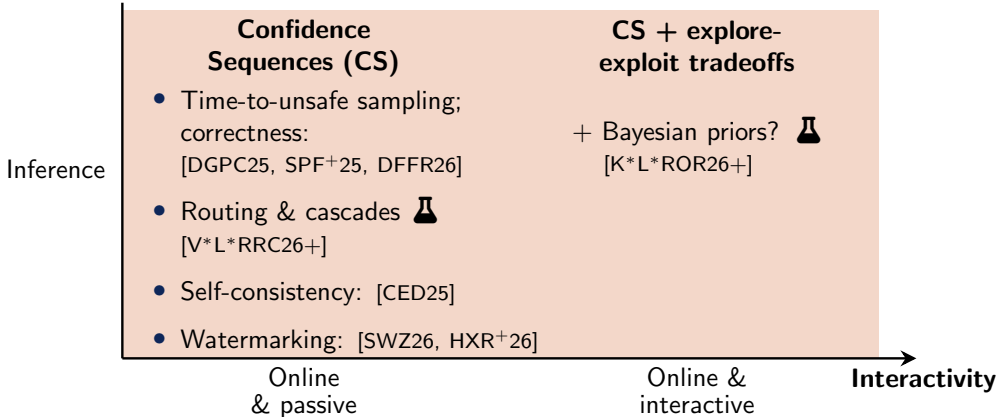
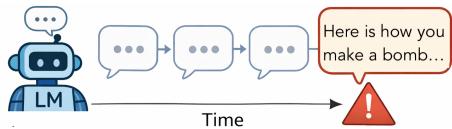
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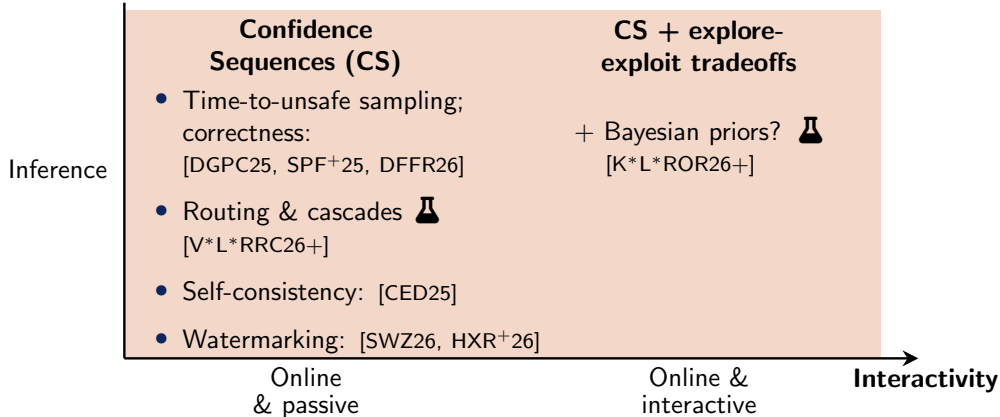


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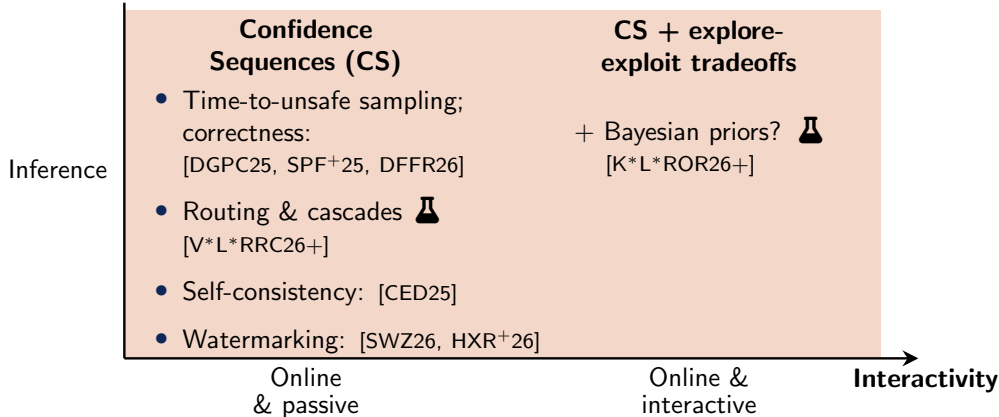
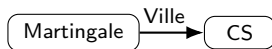
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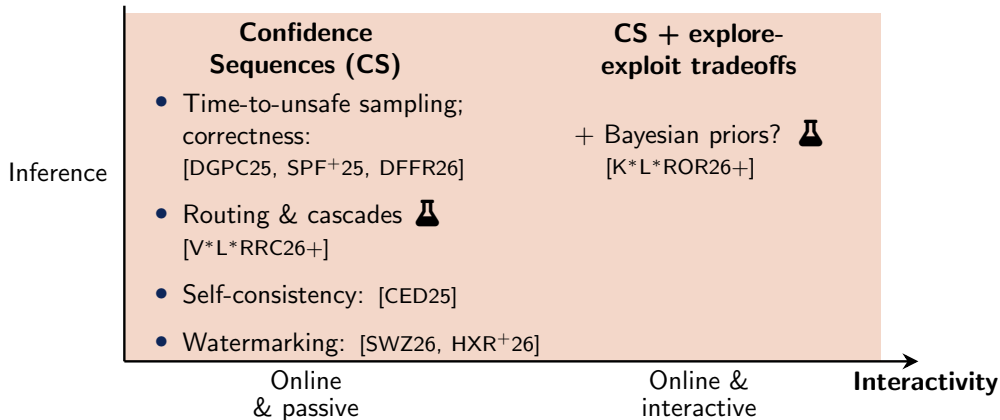
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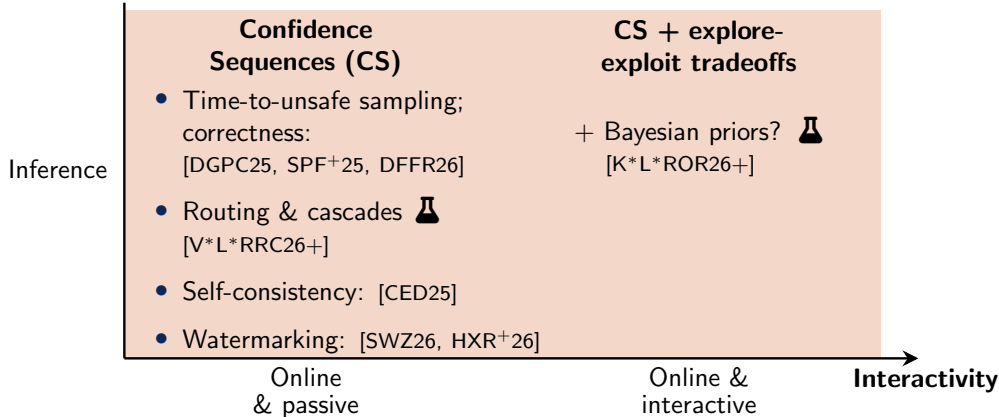
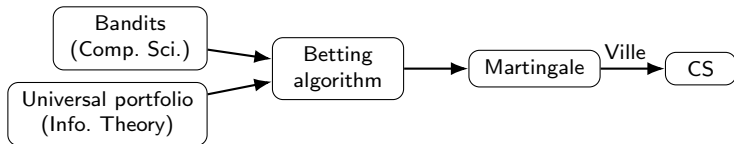
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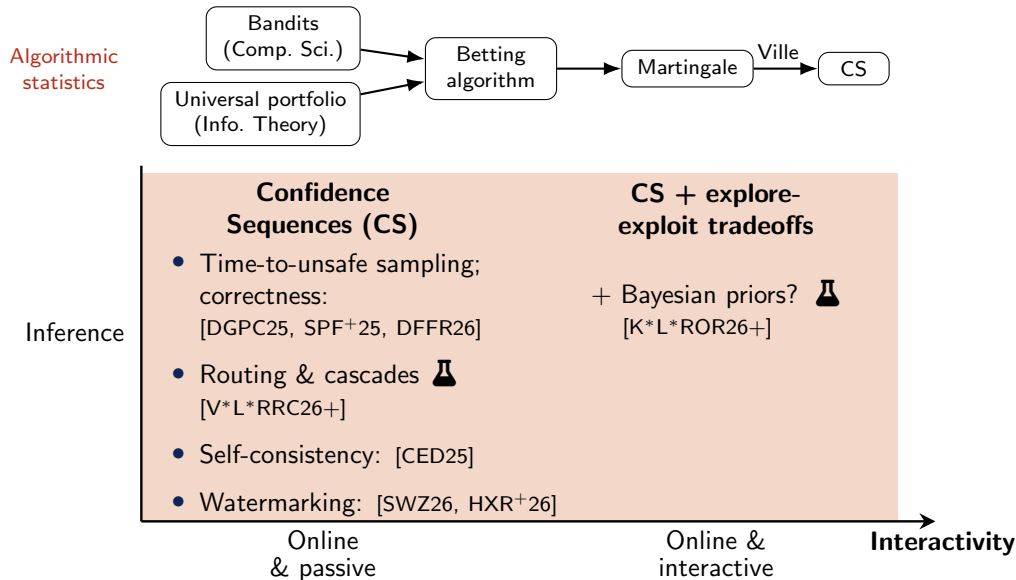
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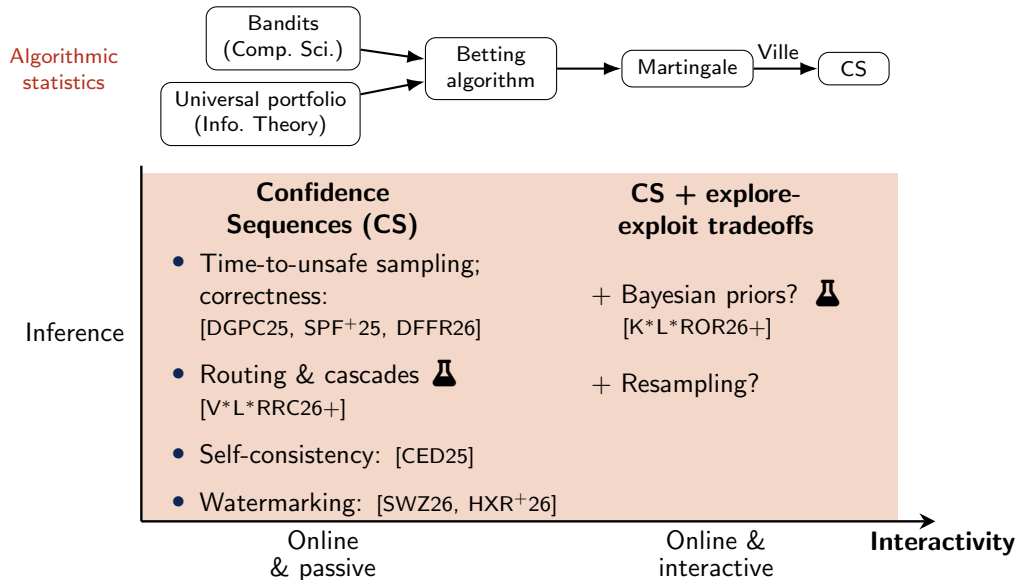
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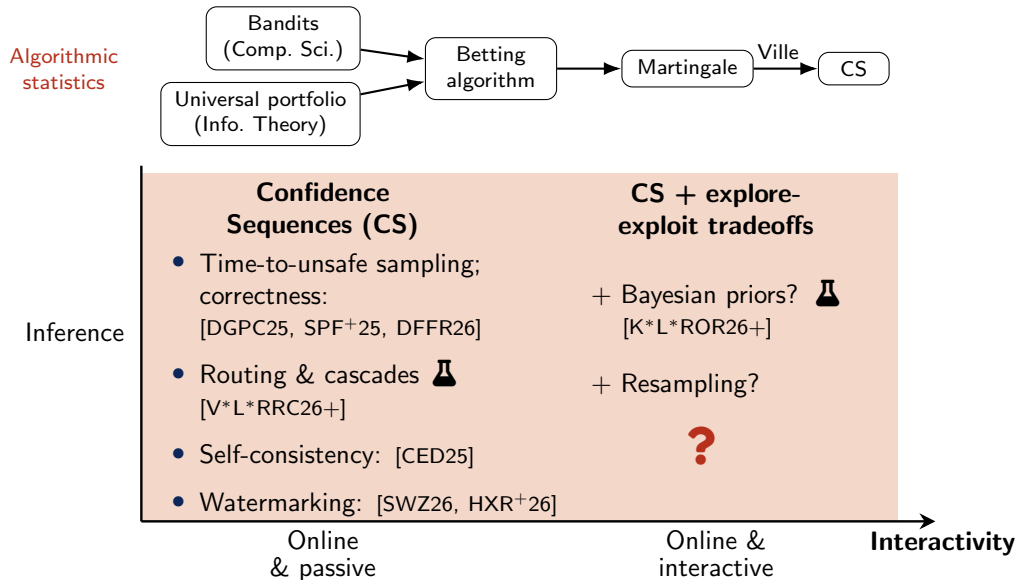
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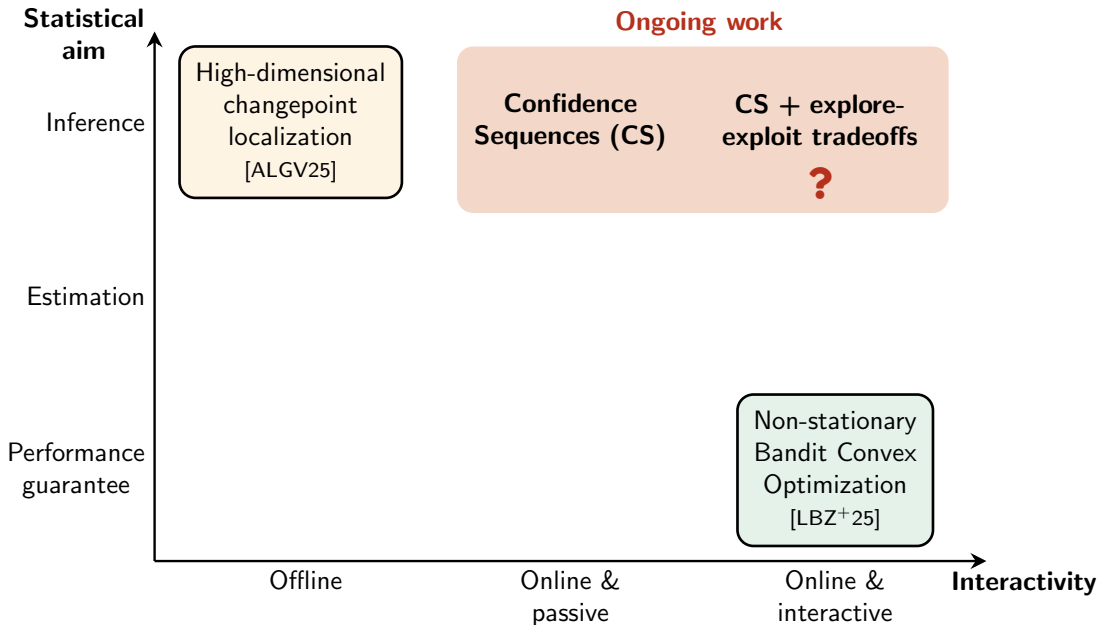
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Wonderful collaborators ★

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- Ramji Venkataramanan, Julia Gontarek (Cambridge)
- Gabriel Arpino (Harvard)
- Patrick Rebeschini, Arya Akhavan (Oxford)
- Dorian Baudry (INRIA)
- Julian Zimmert (Google Research)



References I

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Gabriel Arpino, Xiaoqi Liu, Julia Gontarek, and Ramji Venkataramanan.

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$$W_t = \sum_{j=1}^{n_t} \eta_j e^{-L_{t,l_j}} \quad \text{and} \quad \frac{\partial \log W_t}{\partial (-L_{t,l_i})} = \frac{\eta_i e^{-L_{t,l_i}}}{\sum_{j=1}^{n_t} \eta_j e^{-L_{t,l_j}}}.$$